Advanced Data Modelling and Form Design

Aim

To extend the **Event Management System** in Zoho Creator by implementing advanced field types and features including lookup fields, image/file uploads, subforms, audio/video media, numeric/formula validations, rich text, and digital signatures across two interconnected forms.

Theory

Advanced data modelling in Zoho Creator involves using specialized field types and advanced form components to collect, manage, and validate detailed data. In this system, two forms work together to organize information:

1. Events Form: Manages core event logistics, schedules, and structural data.
2. Event Registration Form: Manages participant details, linking directly to the specific event via a lookup field.

Specialized data structures like lookup fields, image fields, file uploads, subforms, currency metrics, rich text terms, and digital signature blocks are distributed across these forms to build a comprehensive system.

Form 1: Events Form

This form establishes the primary event details and serves as the master record.

- Section A — Event Details & Scheduling
    - Event Name (Single Line field)
    - Event Date & Time (Date-Time field)
    - Event Banner / Thumbnail (Image field)
    - Event Venue Location (Single Line / Address field)
- Section B — Program & Media Assets
    - Event Agenda / Schedule (Rich Text field)
    - Promotional Teaser Video (Video field, optional)
    - Official Venue Blueprint (File Upload field)

    - Event Website URL (URL field)
- Section C — Capacity & Fees
    - Max Capacity (Number field)
    - Base Ticket Fee (Currency field)
    - Early Bird Discount Rate (Percent field)
    - Actual Ticket Price (Formula field, auto-computed: Base Ticket Fee - (Base Ticket Fee * Early Bird Discount Rate))
- Section D — System Identifiers
    - Event ID (Auto Number field, system-generated)
    - Event Asset QR Code (QR/Barcode field, system-generated from Event ID)

Form 2: Event Registration Form

This form captures detailed participant records and hooks dynamically into the Events Form.

- Section E — Attendee Identity & Verification
    - Select Event (Lookup field, auto-fills event details from the master Events Form)
    - Attendee Full Name (Single Line field)
    - Attendee Photo (Image field)
    - ID Proof / Document Upload (File Upload field)
- Section F — Group Registration & Preferences
    - Co-Registrants / Team Details (SubForm field, repeating rows)
        - *Inside SubForm:* Guest Name (Single Line), Guest Email (Email), Food Preference (Dropdown)
    - Company Portfolio URL (URL field, for corporate attendees)
    - Attendee Intro Voice Note (Audio field, optional)

- Section G — Accommodation & Payment Status
    - Requires Accommodation (Decision Box / Checkbox)
    - Registration Fee Paid (Currency field)
    - Payment Receipt Upload (File Upload field)
- Section H — Legal Declaration & Confirmation
    - Terms and Liability Display (Rich Text field, read-only reference block)
    - Attendee Digital Signature (Signature field)
    - Registration Timestamp (Date-Time field, auto-captured upon submission)
    - Registration Ticket ID (Auto Number field, system-generated)

Procedure

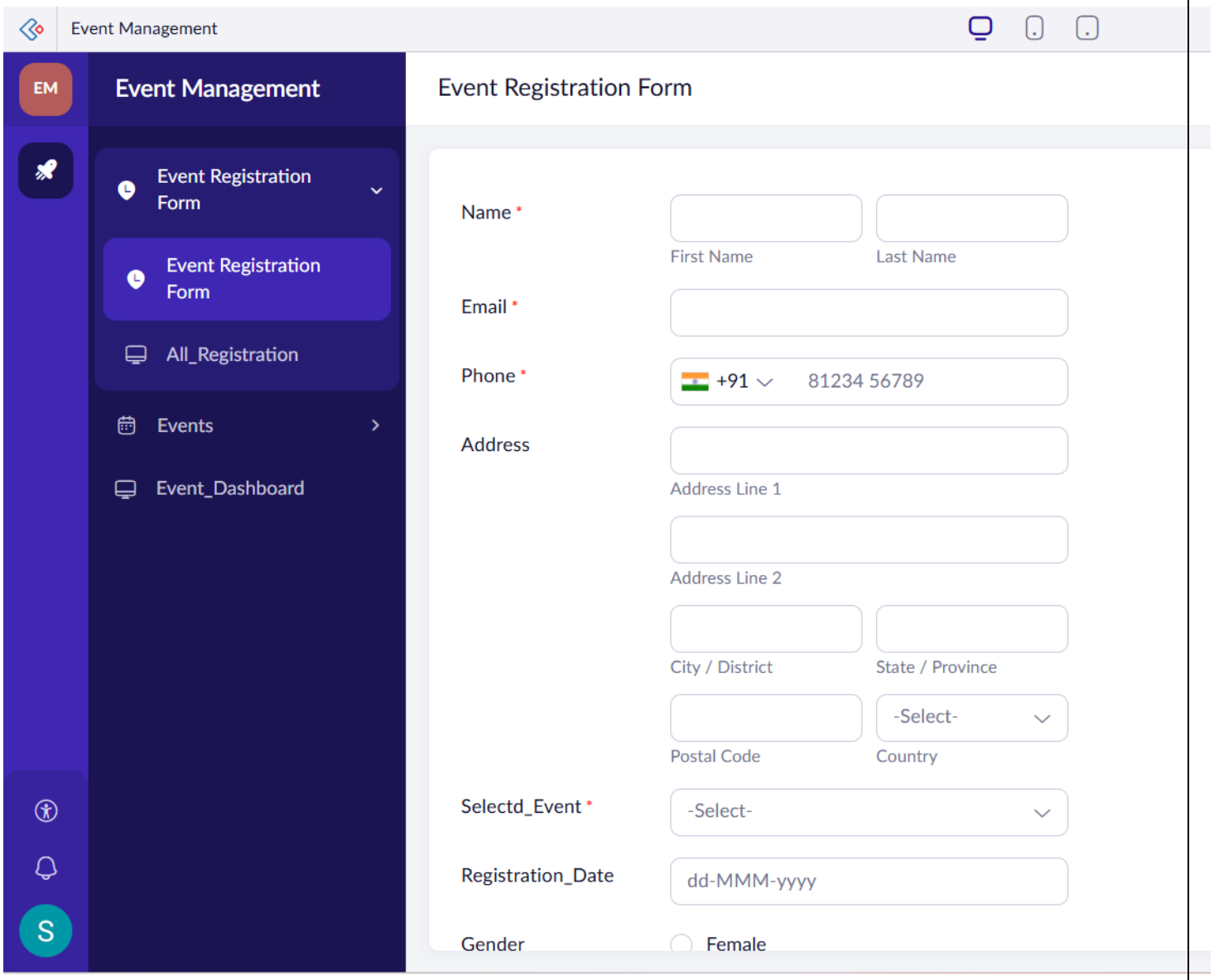
1. Open your workspace application in Zoho Creator.
2. Create Form 1 (Events): Add name, schedule, description, media fields, and fee currencies.
3. Configure the Event ID and QR Code generation logic in the Events form properties.
4. Create Form 2 (Event Registration): Set up basic identity fields.
5. Drag and drop a Lookup field into the Registration Form and link it directly to the Event Name field of the Events Form.
6. Create Section F and add a SubForm to capture multiple accompanying member details per registration.
7. Configure inside the SubForm: Member Name, Member Email, and Member Role.
8. Add the requested media inputs (image uploads, URLs, audio notes) to their respective fields.
9. Create Section G (Fee & Accommodation), adding the Requires Accommodation decision box and registration payment logs.
10. Create Section H (Declaration), adding the terms layout and the Digital Signature element.

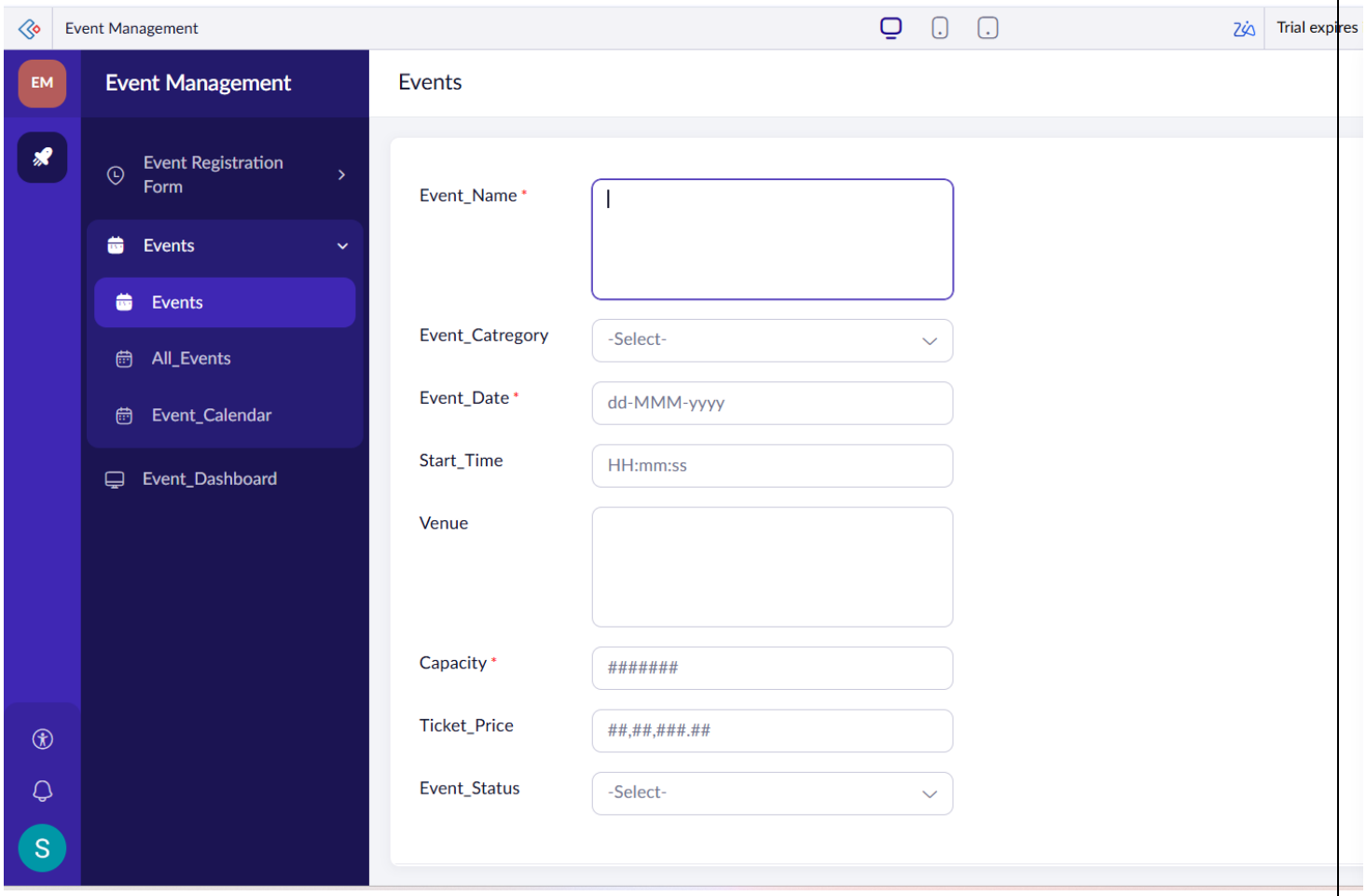
11. Save, publish, and open the application preview.

12. Populate the Events Form first with sample event logistics data to establish the master records.

13. Access the Event Registration Form, choose an active event via the Lookup dropdown, fill out student/attendee information, interact with the SubForm rows, sign digitally, and click Submit.

14. Navigate to the All_Events and All_Registration summary views to verify that the related metrics match correctly.





Result

The Event Management System was successfully built using a two-form architecture in Zoho Creator. The data relationship between the events registry and live attendee records functions correctly via lookup dependencies, validating subform nesting arrays, rich media formatting, and structural validation checks.